



SCHOOL OF COMPUTING AND ENGINEERING SCIENCES
BACHELOR OF SCIENCE IN COMPUTER NETWORKS AND CYBER SECURITY
COURSE OUTLINE

CNS 3106: Wireless Communications

Lecturer: K. Victor Rop **Phone:** 0723-891358 vrop@strathmore.edu
Assistant: Charles Mutiso cmutiso@strathmore.edu
Consultation: Wednesdays **Module Leader:**

Aim: To enable the student to learn the physical properties of waves and signal propagation in air and free space.

Intended Learning Outcomes:

At the end of this course, the learner should be able to

1. Apply the radio frequency spectrum map for frequency allocation to different wireless telecommunication applications.
2. Understand different antennae designs and their applications.
3. Design multiplexing schemes to achieve efficient utilization of radio frequency.

Contact Hours: 45

Prerequisite:

- CNS 2105 - Data and Voice Communications

Lecture time: ***Group A:** Monday 3:15-5:15 & Tuesday 8:15-10:15*
 ***Group B:** Monday 10:15-12:15 & Tuesday 10:15-12:15*

Content, Outcomes and Activities				
Week	Topic	Intended Learning Outcomes	Learning Activities	Assessment/Exercises
1	Introduction to Wireless Communications	At the end of this topic, students will be able to: - Explain the basic concepts and importance of wireless communication systems - Identify key components of wireless communication architecture Overall ILO: <i>To understand the role and structure of wireless communication systems</i>	- Applications of wireless communication (mobile systems, broadcasting, satellite, IoT) - Wireless Communication Architecture (transmitter, channel, receiver) - Introduction to the radio spectrum and its allocation	- Review of application of Wireless Communication - Compare different wireless applications
2 & 3	Transmission and Radio Wave Propagation	At the end of this topic, students will be able to: - Describe different types of radio wave propagation - Analyze atmospheric effects on signals - Understand transmission and reception of signals Overall ILO: <i>To analyze how radio waves propagate in different environments</i>	- Radio spectrum classification (LF, MF, HF, VHF, UHF, SHF) - Types of propagation: surface waves, space waves, sky waves - Reflection, refraction, diffraction mechanisms - Atmospheric effects: fading, attenuation, noise, interference - Basic transmitter and receiver concepts	- Solve propagation-related numerical problems - Calculate path loss under different conditions - Lab/simulation: visualize wave propagation behavior
4	Antenna Fundamentals	At the end of this topic, students will be able to: - Explain key antenna parameters - Interpret radiation characteristics - Differentiate between different types of antennas	- Antenna concepts and functions - Antenna parameters: impedance, radiation resistance, efficiency - Radiation pattern (main lobe, side lobes, beamwidth) - Antenna gain and directivity - Introduction to antenna types (dipole, loop, Yagi, patch, parabolic)	- Plot and interpret radiation patterns - Lab/simulation: antenna radiation pattern visualization

		Overall ILO: <i>To evaluate antenna performance using key parameters</i>		
5	Link Budget & Signal Strength	At the end of this topic, students will be able to: - Apply Friis equation - Perform link budget Overall ILO: <i>To evaluate communication link performance quantitatively</i>	- Concept of link budget - Friis Transmission Equation (derivation and application) - Signal strength factors (distance, frequency, antenna gain) - Link budget calculations	- Case study: Evaluate feasibility of a communication link
6	Review and CAT 1			
7&8	Cellular Communications	At the end of this topic, students will be able to: - Explain cellular system design - Analyze frequency reuse and mobility management Overall ILO: <i>To understand cellular network operations</i>	- Cellular system architecture (base station, MSC, cells) - Frequency reuse and cluster design - Handoff (handover) techniques - Cellular generations (2G–5G overview)	- Analyze handoff scenarios - Case study: mobile network design
9	Satellite Communication Systems	At the end of this topic, students will be able to: - Describe satellite systems - Analyze satellite links Overall ILO: <i>To evaluate satellite communication performance</i>	- Types of satellites (LEO, MEO, GEO) - Orbital characteristics and coverage - EIRP and link budget in satellite systems - Applications (TV broadcasting, GPS, internet)	- Design a basic satellite communication link
10	Multiple Access Techniques	At the end of this topic, students will be able to: - Compare access techniques - Select appropriate schemes	- FDMA, TDMA, CDMA principles - OFDMA and WDMA basics - Random access methods: ALOHA - Demand-based schemes: DAMA, FAMA - Comparison of efficiency and applications	- Solving channel allocation problems - Simulate FDMA, TDMA, CDMA

		Overall ILO: <i>To analyze channel sharing techniques</i>		
11	Review and CAT 2			
12&13	Short-Range Wireless Technologies	At the end of this topic, students will be able to: - Describe short-range technologies - Compare their performance ... Overall ILO: <i>To evaluate and select wireless technologies</i>	- WiFi (IEEE 802.11 standards, architecture) - Bluetooth (architecture, applications) - Zigbee (low-power networks) - Infrared communication - WiMAX overview - Comparative analysis (range, data rate, power, applications)	- Mini-project: design a short-range wireless system (e.g., smart home) - Group presentation of solution

Course Delivery Methodology

1. **Lectures** will be used to introduce material on the formal aspects of the unit. Course materials will be made available digitally.
2. Class discussions, group work, video, online class group discussions, projects

Academic Assessment

Type	Weighting (%)
Examination	60
Coursework	40
CAT 1 & 2	15
Assignments & Quizzes	10
Group Work/Labs	15
Total	100 %

Core Reading Materials

1. Electronic Communications, Dennis Roddy and John Coolen, Prentice Hall 4th ed. (1995) ISBN-10: 013312083X ISBN-13: 978-0133120837
2. Principles of Communication Satellites, Gary D. Gordon and Walter L. Morgan, Wiley-Blackwell, (1993) ISBN-10: 047155796X ISBN-13: 978-0471557968
3. Electronic Communication Systems, Davis Kennedy, McGraw-Hill, (1992) ISBN-10: 0071126724 ISBN-13: 978-0071126724

E-Sources

1. Wireless Communication Tutorial by Tutorials Point.
https://www.tutorialspoint.com/wireless_communication/index.htm
2. Wireless communication tutorial by Naveen T. :
https://www.academia.edu/37296693/Wireless_communication_tutorial
3. Introduction to Wireless Communication Systems: <http://www.wideskills.com/wireless-concepts/introduction-to-wireless-communication-systems>

Assignments/Homework

Students will be given assignments during the semester most of which will be part of the semester project which counts as part of coursework evaluation.

Location and Time

Please refer to the official timetable. Lecture rooms may change from time to time and it is important to keep track of communication from the module leader.

Policies

Punctuality is fundamental. Active **participation** in class discussions is required. It is prohibited to sign the attendance register on behalf of colleagues who are not present.

Plagiarism is a serious offence. If detected in any form in course work and assignments, the following will apply:

- a. In partial or non-serious cases (such as not citing whole word-for-word quotes), half the total possible marks of the assignment are duly struck off.

- b. In serious cases (such as whole duplication of a paper), a zero policy will apply i.e., all offending assignments will be awarded a mark of zero.

Note: The level of seriousness referred to above is at the discretion of the lecturer. Appeals are certainly possible through the relevant channels

Communication Channels

- ✓ Let us Maintain formality
 - e-mail
 - Unit/Module leaders
- ✓ e-learning platform (**WC_26**)